



★ PROBABILITY ★



Name: _____ Date: _____

★ LEVEL 4: EXPERIMENTAL PROBABILITY ★

1 COIN TOSS RESULTS.



A coin was tossed 12 times.
Here are the results.

Result	Number of times
Heads	7
Tails	5

- a) Which happened more often? _____
- b) What is the experimental probability of heads? _____
- c) What is the experimental probability of tails? _____

Remember: total tosses = 12

2 SPINNER TEST.



A spinner with 3 colours was spun 20 times.
Here are the results.

Colour	Number of times
Red	9
Blue	7
Yellow	4

- a) Complete the bar chart using the results above.



- b) Which colour occurred most? _____
- c) Which colour occurred least? _____
- d) Experimental probability of red = _____
- e) Experimental probability of blue = _____
- f) Experimental probability of yellow = _____

Remember: total spins = 20

3 BAG EXPERIMENT.



A bag of counters was picked from 15 times. Here are the results.

Shape	Number of times
★ Star	6
♥ Heart	5
● Circle	4

- a) Which shape seems most likely to be picked? _____
- b) Which shape seems least likely to be picked? _____
- c) Experimental probability of star = _____
- d) Experimental probability of heart = _____
- e) Experimental probability of circle = _____

Remember: total picks = 15

4 COMPARE THEORETICAL AND EXPERIMENTAL.



For a fair coin, the theoretical probability of heads is $\frac{1}{2}$ (one half).

In Activity 1, the coin was tossed 12 times and there were 7 heads.

- a) Is the experimental probability of heads ($\frac{7}{12}$) exactly the same as $\frac{1}{2}$? ☐ Yes ☐ No
- b) Is $\frac{7}{12}$ just close to $\frac{1}{2}$? ☐ Yes ☐ No
- c) Challenge! Explain your answer.



Great job! You are becoming a probability superstar!

probability - fractions

level 3

1 spinner probability

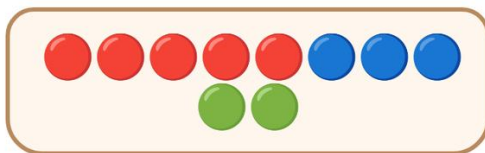


- $P(\text{red}) =$
■ $P(\text{blue}) =$
■ $P(\text{green}) =$
■ $P(\text{yellow}) =$

The spinner has 8 equal sections. Write each probability as a fraction.

2 marbles in a bag

Write each probability as a fraction.



10 marbles in the bag.

- $P(\text{red}) =$
■ $P(\text{blue}) =$
■ $P(\text{green}) =$

3 rolling a die



A fair die is rolled. Write each probability as a fraction (out of 6).

- $P(\text{rolling a } 4) =$
 $P(\text{an even number}) =$
 $P(\text{a number more than } 4) =$

Name: _____

Date: _____



PROBABILITY


LEVEL 1: LIKELY / UNLIKELY


1 Circle likely or unlikely.

1.


 See the sun
in daytime.

likely

unlikely

2.



See a fish fly.

likely

unlikely

3.


 Pick a red apple
from the bowl.

likely

unlikely

4.


 See snow
in a desert.

likely

unlikely

5.


 Roll a number
from 1 to 6
on a die.

likely

unlikely

6.


 Get a cat from
a bag of only
dog pictures.

likely

unlikely

2 Sort the events.

Draw a line from each card to the correct box.

IMPOSSIBLE
UNLIKELY
LIKELY

 The sun comes
out at night.

 It rains on
a rainy day.

 Pick a donut
from a box of
donuts.

 Find two left
socks in a
drawer.

 Your birthday
is in this year.

 See a snowman
in summer.

3 Color the boxes.

 Color the box **GREEN** if the event is likely. Color the box **ORANGE** if the event is unlikely.

1. It will rain tomorrow.


☐

2. You win a prize at school.


☐

3. You find money on the ground.


☐

4. You see a dinosaur in the park.


☐

5. You eat breakfast today.


☐

6. You jump to the moon today.


☐

7. It snows in summer where you live.


☐

8. You draw a heart in class today.


☐

4 Challenge!



Which is more likely:

 picking a **blue** marble from a bag with 5 **blue** and 1 **yellow**,
or picking a yellow marble?

My answer:

Why?

Name: _____

Date: _____



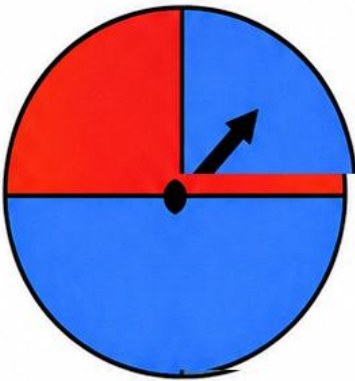
PROBABILITY

- LEVEL 2: SIMPLE PROBABILITY -



1 LOOK AT THE SPINNER.

The spinner has 4 equal sections.



- What color is most likely?

- What color is least likely?

- Is red certain, likely, unlikely, or impossible?

- Is green impossible?

2 ROLL THE DIE.

A fair six-sided die is rolled.



- What is the probability of rolling a 3?
out of 6
- What is the probability of rolling an even number?
out of 6
- What is the probability of rolling a number greater than 4?
out of 6

3 PICK FROM THE BAG.

Look at the shapes in the bag.



- Which shape is most likely?

- Which shape is least likely?

- How many possible shapes are there?

- What is the chance of picking a heart?

out of 10

4 MATCH THE EVENT.

Draw a line to match each event to the correct word.

-  The sun will rise tomorrow.

-  You will find a clover in your math book.

-  You will catch a fish when you go swimming.

-  You roll a 6 on a six-sided die.

-  It will snow on a hot summer day.

-  You will wear shoes to school.

CERTAIN
(Always happens)

LIKELY
(Usually happens)

UNLIKELY
(Does not usually happen)

IMPOSSIBLE
(Cannot happen)